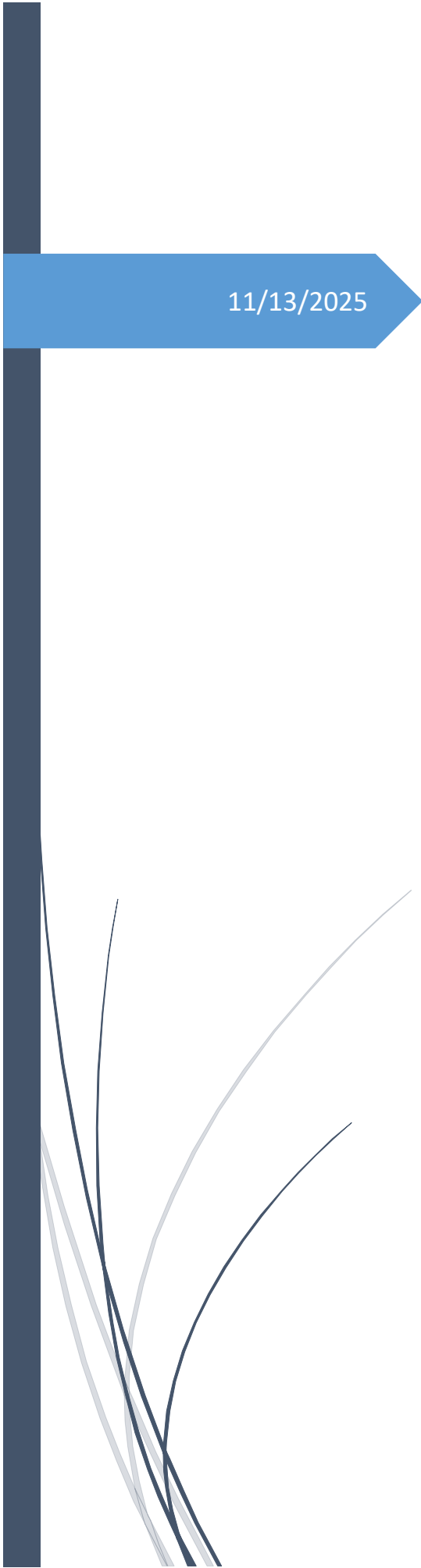




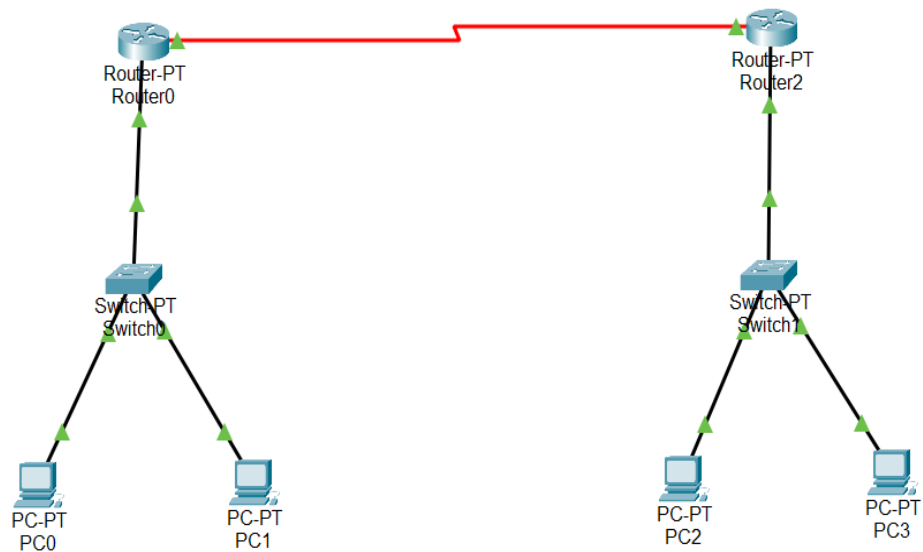
11/13/2025

COMPUTER NETWORK

LAB 13

IFZA RIZWAN
SAP 64281

TOPOLOGY CREATED:



HOW:

Step 1:

Configure all PCs as below 1.

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:F7FF:FE32:E03A

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

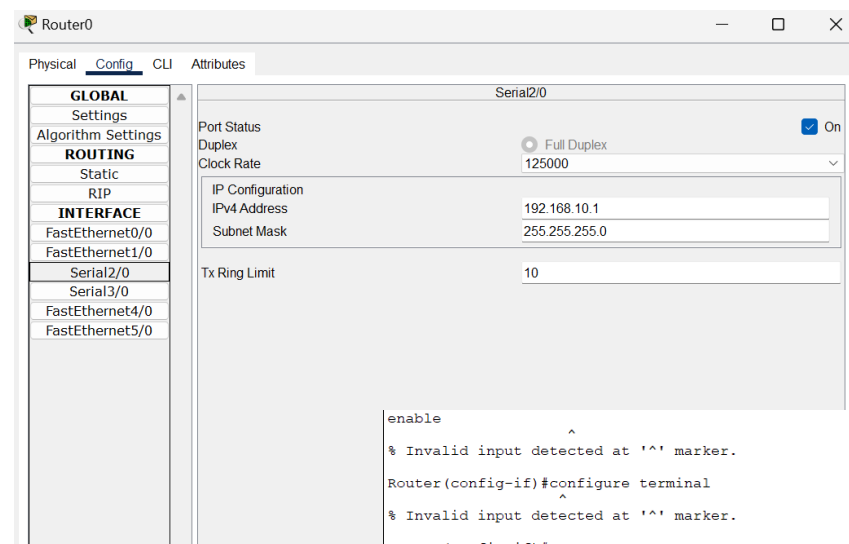
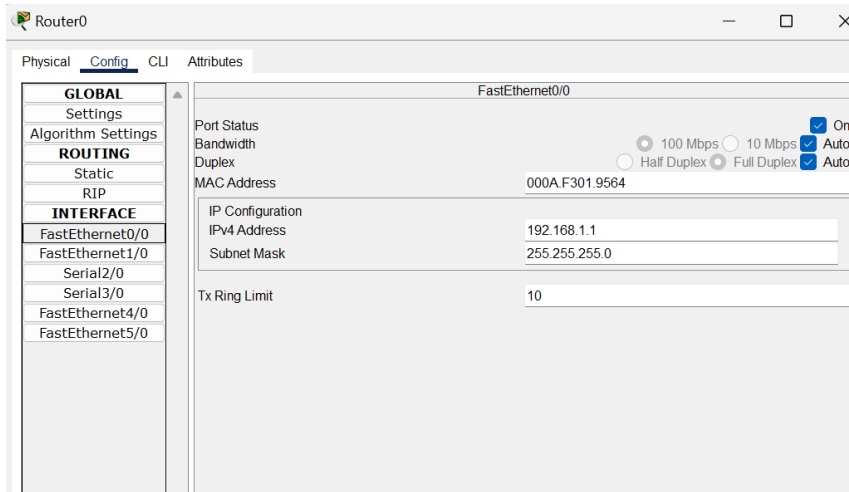
Authentication: MD5

Username:

Password:

Step 2:

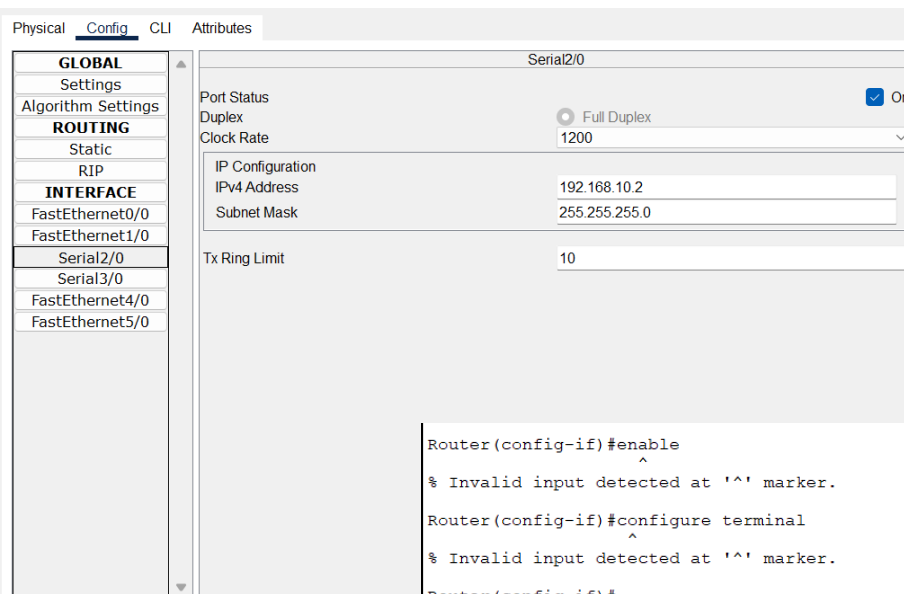
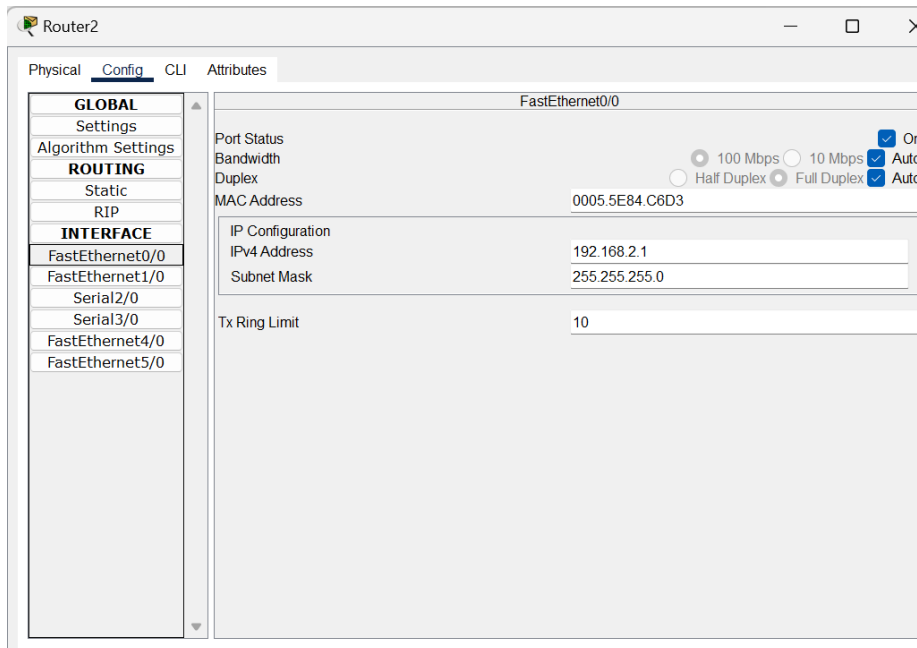
Configure router0:

Configuration of
CLI of router0:

```
enable
^
% Invalid input detected at '^' marker.
Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.
Router(config-if)#
Router(config-if)#interface Fa0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#
Router(config-if)#interface Se2/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#! If this serial interface is connected to the DTE on the other router,
Router(config-if)#! Packet Tracer will show one end as DCE set clock rate on the DCE side:
Router(config-if)#clock rate 64000      ! run this command while still in interface Se2/0 if
this router is DCE
^
% Invalid input detected at '^' marker.
Router(config-if)#exit
Router(config)#
Router(config)#! Add a static route for the remote LAN:
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.10.2
Router(config)#
Router(config)#end
Router#writememory
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```

Configure router1:



Configuration of CLI of router1:

```
Router(config-if)#enable
^
% Invalid input detected at '^' marker.

Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config-if)#
Router(config-if)#interface Fa0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#
Router(config-if)#interface Se2/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#! Only set clock rate here if this end is the DCE cable end:
Router(config-if)#clock rate 64000 ! (only on the DCE end)
^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#
Router(config)#! Static route to left LAN:
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)#
Router(config)#end
Router#writememory
%SYS-5-CONFIG_I: Configured from console by console
```

Ping from pc3 to pc1:

```
PC3
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

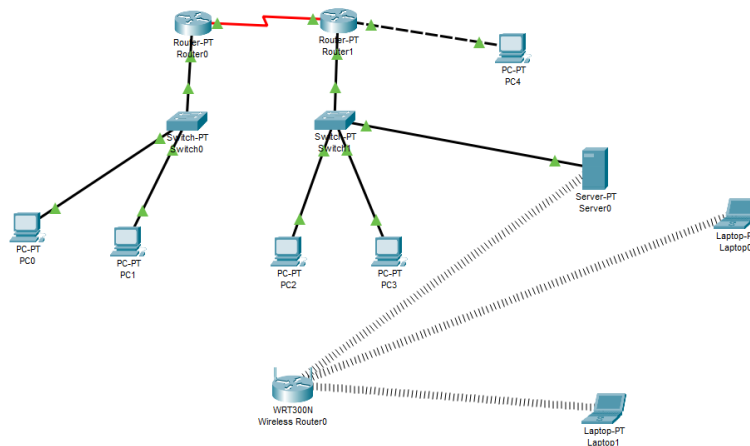
Reply from 192.168.1.3: bytes=32 time=1ms TTL=126
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=12ms TTL=126
Reply from 192.168.1.3: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 6ms

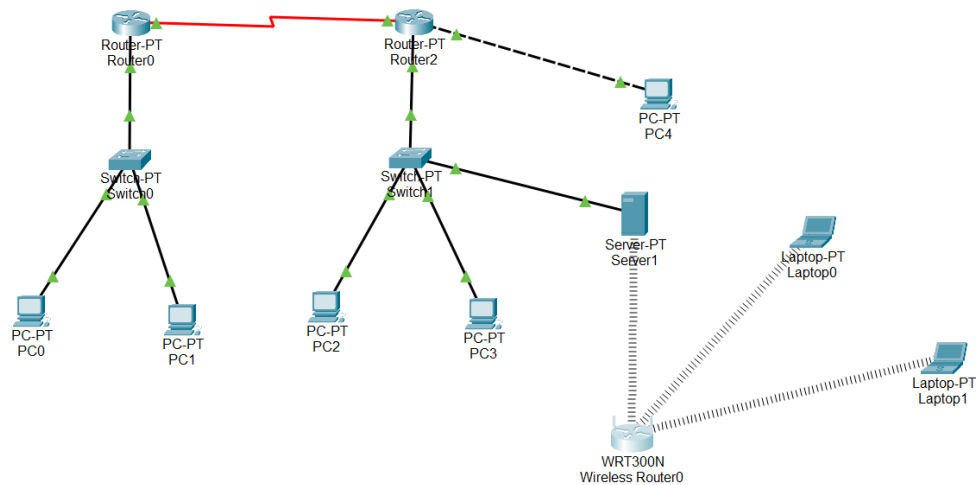
c:\>
```

TASK 2:

Create the topology below using default routing. Take screenshot of each step and show the successful ping of the packets from one pc to another pc.



Topology created:



Half steps are same as above some steps are added:

Step1:

Configure a wireless router.

Wireless Router0

Physical Config **GUI** Attributes

Wireless-N Broadband Router Firmware Version: v0.93.3

Setup	Setup	Wireless	Security	Access Restrictions	Wireless-N Broadband Router	Applications & Gaming	Administration	Status
	Basic Setup		DHCP	MAC Address Clone			Advanced Routing	

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers):

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP: IP Address: 192 . 168 . 0 . 1 Subnet Mask: 255.255.255.252

DHCP Server Settings:

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.0.100

Maximum number of Users: 50

IP Address Range: 192.168.0.100 - 149

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0

Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

WINS: 0 . 0 . 0 . 0

[Help...](#)

Physical Config GUI Attributes

Wireless-N Broadband Router Firmware Version: v0.93.3

Wireless	Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Administration	Status
	Basic Wireless Settings		Wireless Security	Guest Network	Wireless MAC Filter		Advanced Wireless Settings

Basic Wireless Settings

Network Mode: Mixed

Network Name (SSID): ifza wifi

Radio Band: Wide - 40MHz Channel

Wide Channel: 3

Standard Channel: 1 - 2.412GHz

SSID Broadcast: ☒ Enabled ☐ Disabled

[Help...](#)

Physical Config GUI Attributes

Wireless-N Broadband Router Firmware Version: v0.93.3

Wireless	Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Administration	Status
	Basic Wireless Settings		Wireless Security	Guest Network	Wireless MAC Filter		Advanced Wireless Settings

Wireless Security

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: ifza1234

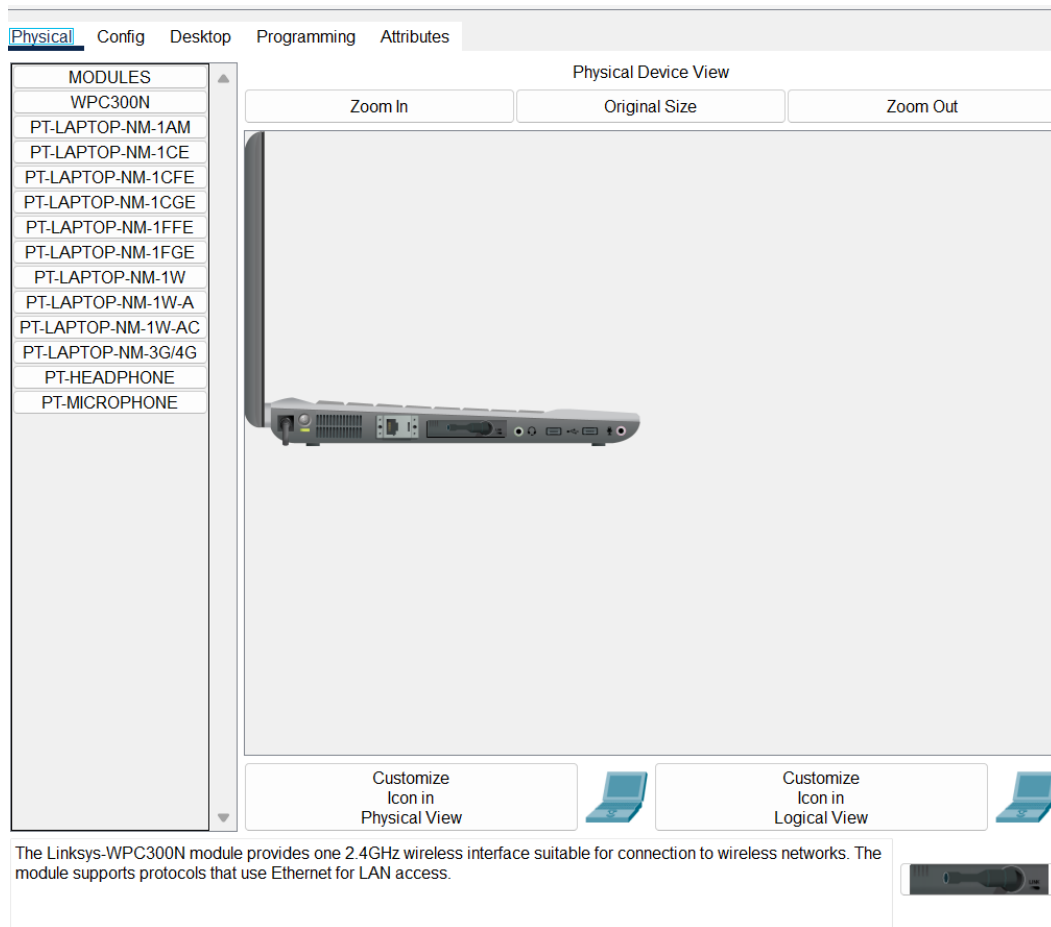
Key Renewal: 3600 seconds

[Help...](#)

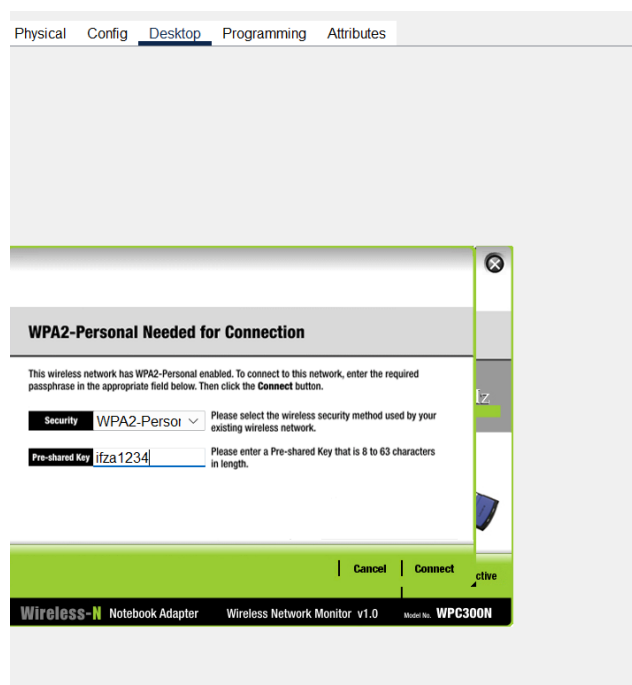
Step2:

Now set laptop for wireless connection.

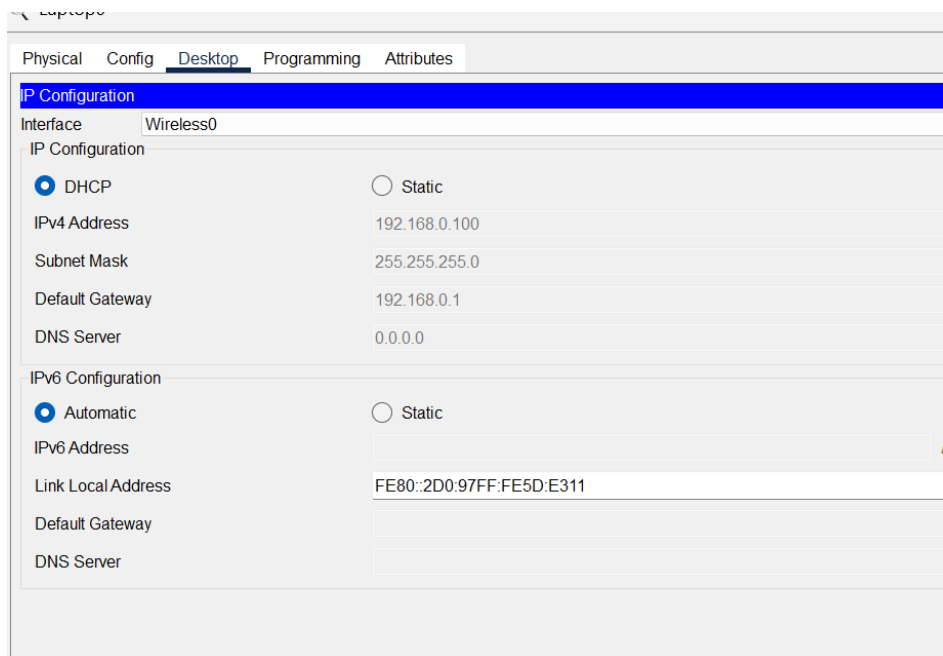
From below pic of physical tab off button of laptop remove vaccum and add WPC300N vaccum and On the laptop.



Now move to desktop->>wireless find network of wireless router I save as ifzawifi shown in tab then try to connect it ask for password add the password that is in the router as in below.

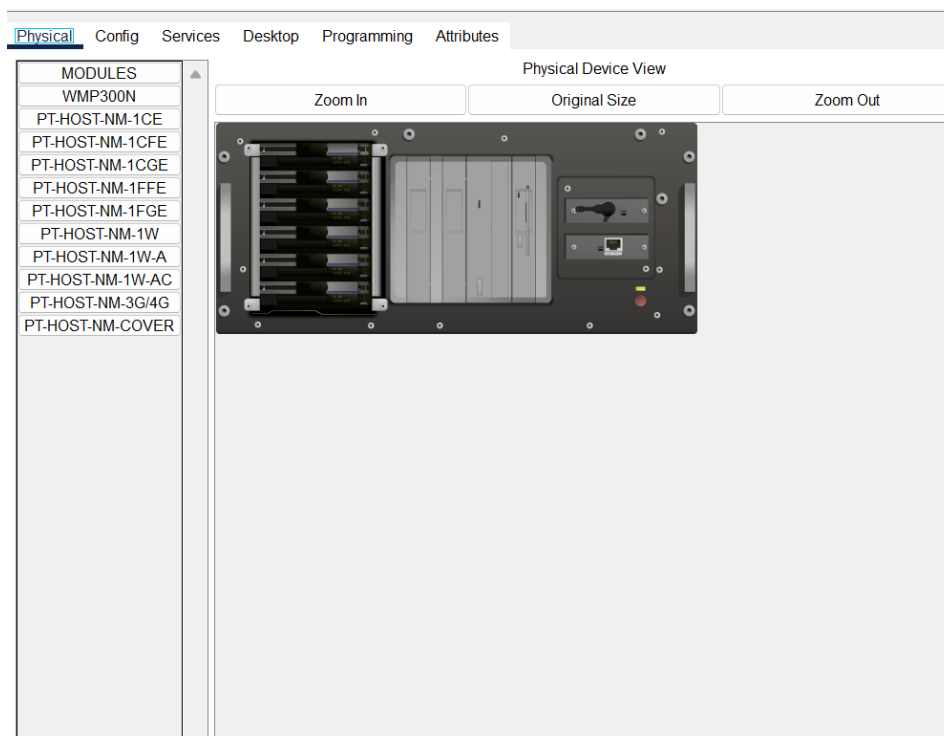


Now get the DHCP ip wirelessly through the router as in below.

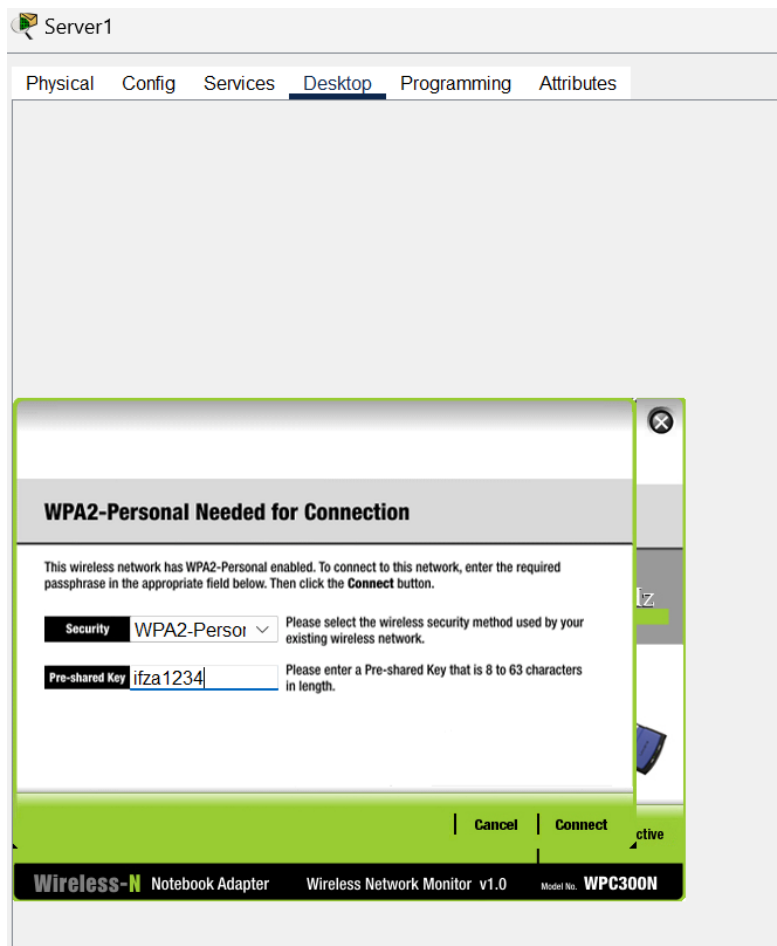


Step 3:

Now configure server so that it can connect in both physically and wirelessly through the wireless router and pt router at the same time. Add 1CFE and WMP300N both in places after turn off the server then again turn on the server.



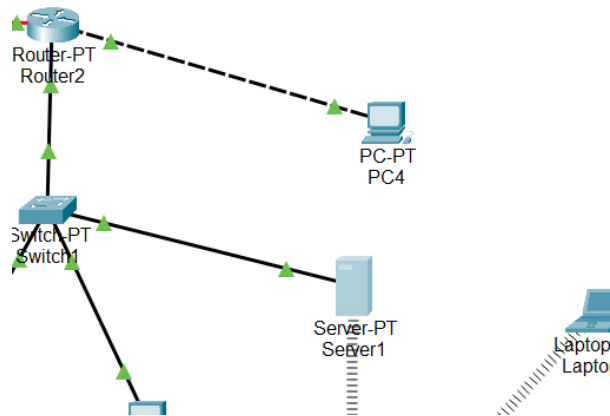
now connect the server wirelessly from desktop->>wireless same as the laptops.



Attain DHCP ip of server via wireless router.

IP Configuration	
Interface	Wireless0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.0.104
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	0.0.0.0
IPv6 Configuration	
☒ Automatic	☐ Static
IPv6 Address	
Link Local Address	FE80::2E0:8FFF:FE62:70E7
Default Gateway	
DNS Server	

Add an additional pc to pt router 2 directly not via switch.



Last step:

Now ping the network for checking the valid working of the network.

PC3

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=36ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=11ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 12ms
C:\>
  
```

PC2

```

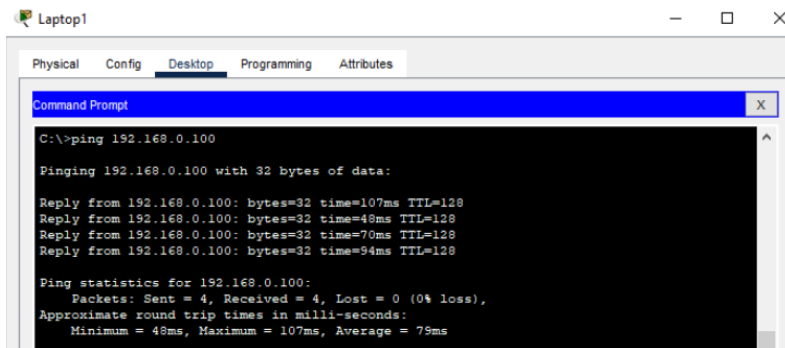
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>
  
```

Pinging laptop0 from laptop1:



The screenshot shows a window titled 'Laptop1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'C:\>ping 192.168.0.100'. The output indicates a successful ping with 4 packets sent and received, 0% loss, and an average round trip time of 79ms.

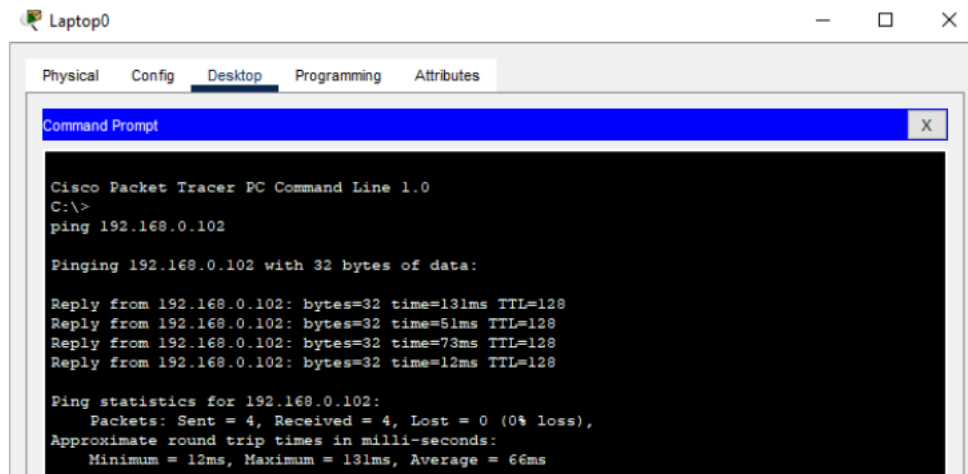
```
C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=107ms TTL=128
Reply from 192.168.0.100: bytes=32 time=48ms TTL=128
Reply from 192.168.0.100: bytes=32 time=70ms TTL=128
Reply from 192.168.0.100: bytes=32 time=94ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 48ms, Maximum = 107ms, Average = 79ms
```

Pinging laptop1 from laptop0



The screenshot shows a window titled 'Laptop0' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'C:\>ping 192.168.0.102'. The output indicates a successful ping with 4 packets sent and received, 0% loss, and an average round trip time of 66ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.0.102

Pinging 192.168.0.102 with 32 bytes of data:

Reply from 192.168.0.102: bytes=32 time=131ms TTL=128
Reply from 192.168.0.102: bytes=32 time=51ms TTL=128
Reply from 192.168.0.102: bytes=32 time=73ms TTL=128
Reply from 192.168.0.102: bytes=32 time=12ms TTL=128

Ping statistics for 192.168.0.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 131ms, Average = 66ms
```

THE END